

[southwestern.edu](https://www.southwestern.edu)

Engagement Opportunities

6–7 minutes

Biological Honors Society - *Beta Beta Beta*

[The Delta Alpha Chapter of Tri-Beta at Southwestern University](#)

provides resources and support to the work of student biologists at Southwestern. First-year students can apply to be Associate Members and then apply for Regular Membership after meeting the criteria of a certain grade point average and completion of the first year sequence of Biology. [Dr. Gonzalez](#) serves as the main faculty mentor for this group, although all the Biology faculty work to support the work of this very active organization. Tri-Beta also contains a subgroup of members focused on pre-health professions including medical, dental and physical therapy and another on graduate school.

Undergraduate Research (on or off-campus)

Many members of the Biology Department have research programs dedicated to conducting research with students, both as a teaching demonstration of scientific skills but also as part of the process of scholarship. The scientific process clearly includes presentation and peer review and we enjoy mentoring our students

in these processes. In terms of writing and critical thinking, collaborative research can result in some combination of the following outcomes:

- Presentation of research (oral or poster presentations) at the [SU RCWS \(see above\)](#) ;
 - Research done with faculty may earn credit hours in the Department of Biology;
 - Presentation (oral or poster) at the [Texas Academy of Science meeting](#) or other regional and/or national scientific meetings; and
 - Eventual publication of a peer-reviewed scientific publication
- Research options occur in different forms depending on the faculty. Some faculty participate in a university-wide program of summer research ([SURF](#) - Summer Undergraduate Research with Faculty) while others create their own tailored arrangements. If interested in the research process, **start a conversation** with individual professors about their research programs:
- [Dr. Erin Borbee](#), Marine Biodiversity and Ecology
 - [Dr. Stacie Brown](#) , Molecular Microbiology
 - [Dr. Romi Burks](#) , Aquatic and Molecular Ecology
 - [Dr. Martín Gonzalez](#) , Molecular Microbiology
 - [Dr. Kim McArthur](#) , Developmental Neuroscience
 - [Dr. Sunny Scobell](#), Physiology
 - [Dr. Max Taub](#), Human Genetics
 - [Dr. Maria Todd](#), Cancer Biology

Off-campus Research (REUs and similar programs)

REU stands for [Research Experiences for Undergraduates](#) , a signature program of the National Science Foundation which offers several opportunities across the country. In addition, a number of medical schools, study-abroad programs, research institutes, and other universities offer REU-like independent research opportunities for students. Students can learn more about these opportunities by talking with faculty or following announcements sent to the Biology list-serve (su-biology@southwestern.edu).

[SU Research and Creative Works Symposium \(SU-RCWS\)](#)

An annual symposium modeled after a typical academic conference where Southwestern students present the results of their research and creative works.

[Study Abroad](#)

The Biology Department encourages as many of its students as possible to study abroad and experience the global nature and reach of science. [The Office of Study Abroad and International Student Services](#) can serve as your first point of contact or you can talk with your academic advisers. The Biology Department Faculty have also assembled lists of programs in Biology and Health Studies to help narrow the vast number of choices to programs that work well with the Southwestern curriculum and our students.

Check them out:

Pre-vetted [Biology Study Abroad Options](#)

Pre-vetted [Health Studies Study Abroad Options](#)

Internships

Students can incorporate internships for credit as part of their Biology degree with the permission of the department. Inquire with [Dr. Romi Burks](#) and check out the additional resources provided by the [Office of Internships](#) within the Center for Integrated Learning.

[Pre-Med and Allied Health Professional Programs](#)

The Pre-Med and Allied Health Committee may have a further list of additional programs that include shadowing medical professions or volunteering with local hospitals. Email [Dr. Kim McArthur](#) and/or check out all the resources available for pre-med, dental, vet, PA, PT, Nurse Practitioner and more on [their website](#).

[King Creativity Grants](#)

Several biology majors have been successful in obtaining [King Creativity Grants](#) to support their interests outside the classroom. The King Creativity Fund empowers students to pursue projects that stretch disciplinary boundaries or combine traditional academic methods with new approaches. In a recent project **Zooxanthellae and *Aliivibrio fischeri*: An Affinity Analysis on the Macro Scale**, Murphy Jacobie '27 set out to discover if coral reef bleaching can be reversed. Jacobie tested to see if the attraction between zooxanthellae, the algae found on corals, and *Aliivibrio fischeri*, a bioluminescent bacteria, could exist on a macroscopic scale. Other microbiologists created a portfolio of

microscope images, [more biologists explored art and science](#) and our environmentalists set out doing [research on the health of the San Gabriel River](#) .

Civic Engagement

Many of our biology students have participated in programs associated with local schools or after school programs. Some biology courses are taught with a community-engaged learning component. More information about these learning opportunities can be found through the [Office of Community-Engaged Learning](#).